## 1.5 Questions to answer (write down your observations)

1. **How many columns does products.csv have? Which ones start with _ (internal columns)?**

- products.csv has 21 columns

```
[5]  print(df.shape)
     ✓ 0.0s

...  (228, 21)
```

- there are 4 internal columns

```
[6]  print(df.dtypes)
     ✓ 0.0s

...  product_id                 str
     product_uuid               str
     brand                      str
     model_name                 str
     colorway                   str
     category                   str
     subcategory                str
     slug                       str
     display_name               str
     price_usd              float64
     _internal_cost_usd     float64
     _supplier_id               str
     _warehouse_location        str
     _internal_cost_code        str
     available_sizes_json       str
     image_urls                 str
     weight_grams             int64
     is_active                 bool
     is_hype_product           bool
     created_at                 str
     tags                       str
     dtype: object
```

2. **How many columns does users.csv have? Can you spot PII (passwords, IPs)?**

users.csv has 28 columns

```
# --- CSV: Read users.csv ---
response = s3.get_object(Bucket="kickz-empire-data", Key="raw/users/users.csv")
df = pd.read_csv(StringIO(response["Body"].read().decode("utf-8")))
```

[14] ✓ 0.4s

```
print(df.shape)
```

[15] ✓ 0.0s

```
(5000, 28)
```

```
print(df.dtypes)
```

[16] ✓ 0.0s

```
user_id                  str
user_uuid                str
email                    str
first_name               str
last_name                str
phone                    str
gender                   str
date_of_birth            str
country_code             str
country_name             str
city                     str
address_line_1           str
address_line_2           str
postal_code            int64
timezone                 str
registration_date        str
is_verified             bool
loyalty_tier             str
newsletter_opt_in       bool
preferred_language       str
_hashed_password         str
_ga_client_id            str
_fbp                     str
_device_fingerprint      str
_last_ip                 str
_failed_login_count    int64
_account_flags         int64
_internal_segment_id   int64
dtype: object
```

3. In orders.csv, what are the possible values for status?

   the possible values for status are delivered, shipped, returned, chargeback, cancelled, cancelled and processing.

```
print(df['status'].unique())
```

[24] ✓ 0.0s

```
<ArrowStringArray>
['delivered', 'shipped', 'returned', 'chargeback', 'cancelled', 'processing']
Length: 6, dtype: str
```

4. In order_line_items.csv, does line_total_usd ≈ unit_price_usd × quantity?

   yes

```
# --- CSV: Read order_line_items.csv ---
response = s3.get_object(Bucket="kickz-empire-data", Key="raw/order_line_items/order_line_items.csv")
df = pd.read_csv(StringIO(response["Body"].read().decode("utf-8")))
```

[25] ✓ 0.9s

```
df['test'] = df['quantity'] * df['unit_price_usd']
print(df['test'].head())
```

[30] ✓ 0.0s

```
0    100.45
1     25.00
2     61.59
3    235.47
4    204.43
Name: test, dtype: float64
```

```
print(df['line_total_usd'].head())
```

[31] ✓ 0.0s

```
0    100.45
1     25.00
2     61.59
3    235.47
4    204.43
Name: line_total_usd, dtype: float64
```

5. In reviews.jsonl, which columns start with _? What do _moderation_score and _sentiment_raw look like?

_moderation_score , _sentiment_raw, _toxicity_score , _language_detected, _review_source

Both _moderation_score and _sentiment_raw are numeric columns (float64), representing model scores or probabilities.

```
print(df_reviews.dtypes)
```

[11] ✓ 0.0s

```
review_id                          str
product_id                         str
product_name                       str
user_id                            str
user_name                          str
rating                           int64
title                              str
body                               str
verified_purchase                 bool
submitted_at          datetime64[us, UTC]
moderation_status                  str
moderated_at          datetime64[us, UTC]
helpful_votes                    int64
reported                          bool
photos_count                     int64
_moderation_score              float64
_sentiment_raw                 float64
_toxicity_score                float64
_language_detected                 str
_review_source                     str
dtype: object
```

6. In the clickstream Parquet file, what does the event_type column contain? What _-columns exist?

The event_type column contains only one value “page_view”, then it tracks page view activity only.

```
df_click['event_type'].unique()
```

[22] ✓ 0.0s

```
<ArrowStringArray>
['pageview']
Length: 1, dtype: str
```

There is 5 internal columns -> _ga_client_id , _gtm_container_id, _dom_interactive_ms, _dom_complete_ms, _ttfb_ms

```
print(df_click.shape)
print(df_click.dtypes)
print(df_click.head())
```

[12] ✓ 0.0s

```
(5000, 27)
event_id                  str
event_type                str
timestamp                 str
timestamp_epoch_ms      int64
user_id                   str
session_id                str
page_url                  str
page_path                 str
page_type                 str
referrer_url              str
referrer_source           str
user_agent_raw            str
ip_address                str
viewport_width          int64
viewport_height         int64
screen_color_depth      int64
device_pixel_ratio    float64
accept_language           str
is_bot                   bool
_ga_client_id             str
_gtm_container_id         str
_dom_interactive_ms     int64
_dom_complete_ms        int64
_ttfb_ms                int64
```